Source-Based AI Recommendation Explanations and Human Decision-Making

Project Status

Independent Research Project — Research Design Complete / Pilot Implementation Pending

Human-participant data have not yet been collected; the analyses described below are planned analyses.

Research Motivation

AI recommendation systems increasingly provide explanations for why particular products, content, or decisions are recommended. However, explanations may differ not only in what they say, but also in **where the supporting information comes from**.

A recommendation may be justified using a user's prior behavior, other consumers' reviews, product attributes, or a combination of multiple information sources. These source disclosures may shape how users interpret the recommendation and whether they trust and rely on the AI system.

This project examines **explanation provenance** as a design feature of AI recommendation systems.

The central question is:

How does the information source disclosed in an AI-generated recommendation explanation influence users' perceptions of relevance, credibility, privacy, trust, and willingness to rely on the recommendation?

Research Design

I developed a **five-condition between-subjects online experiment** using an AI-assisted e-commerce recommendation scenario.

Participants are randomly assigned to one of five recommendation-explanation conditions:

1. **No-source control**
2. **Behavior-based explanation**
3. **Review-based explanation**
4. **Product-feature-based explanation**
5. **Mixed-source explanation**

Across conditions, participants receive the same product recommendation and product information. The primary manipulation is the information source that the AI system identifies as the basis for its recommendation.

The recommended product is a fictional pair of wireless headphones, allowing the experiment to minimize preexisting brand attitudes while maintaining a realistic consumer-decision context.

Conceptual Model

The project proposes that different explanation sources operate through distinct psychological mechanisms.

Behavior-Based Explanations

Behavior-based explanations may increase **perceived relevance** because they explicitly connect the recommendation to the user's previous browsing or shopping activity.

At the same time, revealing the use of behavioral data may increase **privacy concern**.

This produces a theoretically important tension:

Behavioral personalization → greater relevance → greater trust

while simultaneously:

Behavioral personalization → greater privacy concern → lower trust

The net effect of behavioral explanations therefore depends on the balance between personalization benefits and privacy costs.

Review-Based Explanations

Review-based explanations identify other consumers' evaluations as evidence supporting the recommendation.

These explanations are expected to increase **perceived source credibility**, which may subsequently increase trust in the AI recommendation.

Product-Feature-Based Explanations

Product-feature explanations connect the attributes of the recommended item with preferences explicitly provided by the user.

This source may increase **perceived relevance** without requiring the system to reveal extensive use of personal behavioral data.

Mixed-Source Explanations

Mixed-source explanations combine behavioral information, customer reviews, and product-feature information.

Rather than assuming that more information is necessarily better, the mixed-source condition is treated as exploratory. Combining sources may strengthen perceived relevance and credibility, but it may also increase privacy concerns or informational complexity.

Key Hypotheses

The experiment evaluates several theoretically motivated predictions.

H1. Behavior-based and product-feature-based explanations will increase perceived relevance relative to a no-source control.

H2. Behavior-based explanations will generate greater situational privacy concern than review-based, product-feature-based, and no-source explanations.

H3. Review-based explanations will increase perceived source credibility relative to a no-source explanation.

H4. Perceived relevance and source credibility will be positively associated with trust, whereas privacy concern will be negatively associated with trust.

H5. Greater trust in the AI recommendation will be associated with greater willingness to rely on the recommendation.

The study further examines whether explanation source affects reliance indirectly through relevance, credibility, privacy concern, and trust.

The mixed-source condition is examined as an exploratory research question rather than being assigned a predetermined directional effect.

Measurement

The study uses multi-item measures of:

- perceived relevance;
- perceived source credibility;
- situational privacy concern;
- trust in the AI recommendation; and
- reliance intention.

Participants also make a behavioral decision between accepting the AI-recommended product and continuing to search for another product.

Separate manipulation checks assess whether participants correctly perceive the intended recommendation-information source.

Planned Analysis

The analysis plan includes:

- manipulation checks;
- scale reliability assessment;
- one-way ANOVA;
- theoretically specified planned contrasts;
- multiple linear regression;
- logistic regression for behavioral reliance;
- mediation/path analysis; and
- bootstrap confidence intervals for indirect effects.

The analysis explicitly distinguishes experimentally manipulated condition effects from associations among measured psychological constructs.

Mixed-source comparisons are treated as exploratory.

Power Analysis

An a priori power analysis was developed for the five-condition between-subjects experiment.

Using an initial design-stage assumption of:

- $\alpha = .05$;
- power = .80;
- five experimental groups; and
- Cohen's f = .20,

the minimum balanced analyzable sample is approximately **N = 305**.

The current main-study recruitment target is approximately **N = 360**, or roughly 72 participants per condition, allowing for exclusions and incomplete responses.

The effect-size assumption is treated as a planning assumption and may be updated if pilot evidence or stronger prior empirical estimates become available.

Pilot Plan

Before formal data collection, the study will undergo two pilot stages.

Stage 1 — Internal Functionality Testing

A small number of testers will evaluate:

- instructions;
- experimental flow;
- random assignment;
- stimulus presentation;
- response recording; and
- general usability.

Stage 2 — Manipulation and Measurement Pilot

A small independent pilot sample will be used to examine:

- whether participants distinguish the five source conditions;
- whether stimulus wording is sufficiently matched;
- whether the measurement scales perform reasonably;
- whether overlapping constructs remain empirically distinguishable;
- whether the behavioral choice produces meaningful variation; and
- whether the study can be completed without substantial confusion or fatigue.

Pilot findings will primarily inform experimental refinement rather than substantive theoretical conclusions.

Research Contribution

The project shifts attention from the general question of whether AI explanations increase trust to a more specific question:

What happens when an AI system tells users where the information supporting its recommendation came from?

This distinction matters because different information sources may create different benefits and risks.

A source that improves personalization may simultaneously create privacy concerns. A source that appears credible may increase trust without necessarily improving the quality of the user's decision. Combining multiple sources may increase informational richness without guaranteeing better reliance.

The project therefore contributes to research on:

- human–AI decision-making;
- explainable recommendation systems;
- information provenance;
- trust in AI;
- privacy and personalization; and
- human-centered design of AI decision support.

Relationship to My Broader Research Agenda

This project is part of a broader research interest in how the **provenance and verifiability of AI-generated information** shape human judgment.

My related project, **Verification Theater: Source Verifiability and Trust in AI Recommendations**, examines whether citation-like interface cues can create trust when underlying evidence is not meaningfully traceable.

Together, the two projects address complementary questions:

Verification Theater:
Can users actually verify the evidence behind AI-generated claims?

Source-Based Explanations:
How does the type of information source supporting an AI recommendation shape users' perceptions and reliance?

Together, these projects motivate a broader research agenda on designing AI systems that support not merely greater trust, but better-calibrated and more informed human reliance.

Current Status and Next Steps

Completed

- conceptual model;
- five-condition experimental design;
- matched experimental stimuli;
- manipulation checks;
- hypotheses;
- measurement framework;
- experiment specification;
- analysis plan;
- a priori power analysis;
- experiment flow; and
- pilot protocol.

Next

- implement the pilot experiment on an appropriate experimental platform;
- conduct internal functionality testing;
- evaluate manipulation and measurement performance;
- refine the experiment based on pilot evidence; and
- prepare the finalized study for preregistration and formal data collection.